What could externalising a value mean? Four mechanisms, four verdicts

EXCHANGE

writing frees the workspace

prediction. a written value leaves the workspace until it is reused

NOTHING TO FREE

masked = severed in three models (p≥0.30). no stored copy exists

DUPLICATION

the page is a redundant backup

prediction. a written value is also held internally across the gap

REJECTED

whole-transcript mask falls to the 10% floor. the value lives on the page

COMPLEMENTARY

page and workspace hold different content

prediction. CoT values are absent from the direct-mode readout

REJECTED

values surface internally, in computation order, in both arms (3 seeds)

EPIPHENOMENAL

CoT helps via extra compute, not content

prediction. matched-length filler rescues the ablated model

REJECTED

own reasoning 98%, best filler = no prefix (p=0.79, 3 types, 3 seeds)